LEVIATHAN ENGINE 2

Macro Shape Chain Map Specification

Forward-Mapping DXY Price Action via Institutional Pressure Zone Anchored Macro Shape Chains

Version 1.1
May 2026

Part of the Leviathan Ultra-Quant Agent Architecture
Engine 1: DXY Institutional Pressure Zone Map (9-Pillar 6D Hybrid System) v4.3
Engine 2: Macro Shape Chain Map (This Document)

Report Instrument Scope: DXY | AUD/USD (AU) | GBP/USD (GU) | EUR/USD (EU)

1. Architectural Overview

The Leviathan Ultra-Quant Agent operates on a dual-engine architecture. **Engine 1** (the existing 9-Pillar 6D Hybrid System v4.3) maps Institutional Pressure (IP) zones across the DXY landscape, identifying exact price levels where institutional orderflow, option positioning, mathematical structures, and structural extremes converge. Engine 1 answers the question: Where are the walls?

Engine 2 (specified in this document) is the Macro Shape Chain Map. It forward-maps the shape of the coming week's price action by identifying which macro shape chain will dominate, anchored to the IP zones produced by Engine 1. Engine 2 answers the question: What path will price take through those walls?

The critical insight driving Engine 2 is that macro shape chains are not random. Over 8 months of backtesting, recurring shape chains have been identified with measurable frequency distributions. More importantly, the IP zones from Engine 1 function as **anchor points** for each macro shape: they define the consolidation range for Swing Scalps, the zone whose failure triggers a Breakout Surge, the level whose rejection produces a Reverse Whiplash, and the magnet target for Piercing variants. Without IP zones, macro shape prediction is pattern recognition with no structural basis. With IP zones, it becomes a deterministic forward map.

1.1 Instrument Scope

The Leviathan Quant Report covers a fixed instrument universe of four instruments: **DXY** (the primary analysis canvas and the subject of all Engine 1 and Engine 2 mapping), plus **AUD/USD (AU)**, **GBP/USD (GU)**, and **EUR/USD (EU)**. These three component pairs are the only pairs the operator trades alongside DXY. Other DXY component pairs (USD/JPY, USD/CHF, USD/CAD, and SEK-based pairs) may be referenced as supplementary data sources for identifying IP zone levels on DXY, but they do not receive individual pair-level analysis or reporting in the output. Their sole function within the system is to contribute to the discovery and validation of DXY IP zones through cross-pair confluence analysis.

This scoping decision has practical implications for the report output. The Chain Map section includes shape chain analysis for DXY (primary) and cross-pair confluence signals from AU, GU, and EU. No independent shape chain maps are produced for non-traded pairs. Similarly, the Risk Flag section flags cross-pair alignment only among the traded pairs. When non-traded pairs show alignment or divergence with the DXY shape chain, this is noted as supplementary context but does not trigger Risk Flags or Decision Zone modifications on its own.

Dimension	Engine 1 (IP Zones)	Engine 2 (Shape Chains)
Question Answered	Where are the institutional walls?	What path does price take through them?
Output	Zone map with levels, scores, failure probabilities	Shape chain sequence with IP-anchored transitions

Time Orientation	Static map as of Friday close	Dynamic forward map for the trading week
Data Source	9 Pillars across 6 dimensions	IP zone map + macro shape taxonomy + chain frequency data
Validation	11/11 level capture (May 18-22)	8 months of backtested chain observations

Table 1: Engine 1 vs Engine 2 Comparison

2. Macro Shape Taxonomy

Four base macro shapes describe all DXY price action on the Higher Timeframe (HTF: 4H/D1/W1). Every trading session, without exception, exhibits one of these shapes or a variant thereof. The shapes are not subjective pattern labels; they are mechanically defined by the relationship between price action and the IP zone architecture. The distinction between shapes is determined by how price interacts with the IP zones mapped by Engine 1, not by visual pattern recognition alone.

2.1 Swing Scalps (SS)

Definition: Price oscillates within a defined range bounded by IP zones above (resistance ceiling) and below (support floor), forming a series of swing highs and swing lows that respect these boundaries. Each swing typically spans one to three trading sessions. The range is explicitly defined by Engine 1 IP zones: the upper bound is a Heavy or Extreme pressure resistance zone, and the lower bound is a Heavy or Extreme pressure support zone.

IP Zone Signature: Both the ceiling and floor IP zones have Zone Failure Probability (ZFP) below 30%. The regime classification for both boundary zones is "Consolidation" or "Momentum" (but not "Breakout"). Option expiry positioning (Protocol A) cages price between large strikes that are still active. The option cage effect (NY Cut at 10:00 AM) creates repeated Spring Release bounces that manifest as the individual scalps within the range.

Duration: Swing Scalps lasting more than one full trading session without a breakout attempt are classified as SS. If a recovery phase persists for more than one full session, it is reclassified as SS rather than Recovery. This temporal distinction is critical: Recovery is transient (half-session to one session), while SS is sustained (multi-session consolidation).

Subtype - Double Whiplash Pattern: Within SS, a common micro-pattern occurs where two rapid whiplash moves happen in succession within the range. This is labeled "SS (Double Whiplash)" and indicates heightened volatility within the consolidation range, often preceding a breakout attempt.

2.2 Breakout Surge (BS)

Definition: Price breaks decisively through an IP zone that previously acted as a ceiling (for upside BS) or floor (for downside BS), then surges directionally toward the next IP zone or Structural Extreme Magnet. The breakout is confirmed by a sustained move beyond the zone (15+ minutes, per the Dynamic Discovery Protocol in Engine 1), not merely a wick or brief spike.

IP Zone Signature: The failing zone has ZFP above 45%, with active breakout pre-conditions from Engine 1 Pillar 9: option expiry timing (Spring Release), macro catalyst alignment, increasing momentum, and/or cascade effect from a prior zone failure. The Measured Move Projection from Pillar 9 Step 4 provides the target for the surge. The target is always anchored to the nearest Structural Extreme Magnet (Repeated High Magnet, Confluent Ceiling, etc.) from Pillar 5 Step 0/0.5.

Duration: Typically 1-3 sessions. The initial breakout candle is explosive; the second leg (if present) is often even stronger as institutional algorithms cascade stops. The Breakout Surge ends when price reaches the target IP zone or Structural Extreme Magnet and begins to react.

Variant - Piercing Breakout Surge (PBS): A Breakout Surge that is so powerful it pierces directly through one or more intermediate IP zones without pausing. PBS occurs when the failing zone had extremely high ZFP (above 70%), the Measured Move Projection pointed to a distant Structural Extreme Magnet, and the macro catalyst was exceptionally strong. PBS is the strongest directional move in the taxonomy and almost always produces a Reverse Whiplash or Whiplash Storm at its termination point.

2.3 Recovery (R)

Definition: A temporary stabilization or mild retracement that occurs immediately after a Breakout Surge or Whiplash. Price consolidates or slowly drifts back toward the prior IP zone level, but does not fully reverse the preceding move. Recovery is always a transitional shape that bridges two higher-energy shapes.

IP Zone Signature: Recovery typically occurs at a mid-range IP zone (Moderate to Moderate-Heavy pressure) that provides enough friction to slow the prior move but not enough to reverse it. The ZFP of the zone where Recovery occurs is typically 20-40%, meaning it will hold temporarily but is not a structural anchor. Recovery zones are often Post-Expiry Ghosts or One-Touch zones from Engine 1, meaning they have structural memory but lack active institutional defense.

Duration: Half-session to one full session. This is the critical temporal distinction: if stabilization persists for more than one full session, it is reclassified as Swing Scalps. The transition from R to SS represents a structural upgrade where the market has accepted the new range and is no longer in a transitional state.

Variant - Recovery Surge (RS): Instead of a slow consolidation, Recovery takes the form of a sharp directional move back toward the pre-breakout level. This occurs when the prior BS or Whiplash was overextended and institutional flow reverses aggressively. RS is less common than standard Recovery but signals that the prior directional move may have been a false breakout, increasing the probability of a Reverse Whiplash as the next shape.

2.4 Whiplash (WL)

Definition: A rapid, violent directional move that immediately reverses, creating a spike-and-reject pattern. Whiplash occurs when price surges toward an IP zone, briefly pierces it (triggering stops), then is violently rejected back in the opposite direction. The key characteristic is the rejection at the IP zone,

not the initial surge.

IP Zone Signature: Whiplash occurs at IP zones classified as Hard Walls or Standard Walls with ZFP below 20%. These are zones that WILL hold, and the brief pierce is a liquidity sweep (per Engine 1 Pillar 5 Step 0: Liquidity Sweep Target). The market surges toward the zone to trigger stop-loss orders clustered at the prior extreme, then reverses as institutional limit orders defend the level. The Liquidity Sweep Zone designation in Engine 1 is the primary identifier for Whiplash-producing zones.

Variant - Reverse Whiplash (RWL): A Whiplash that occurs in the opposite direction of the preceding Breakout Surge. After an upside BS, a RWL is a violent downside rejection. After a downside BS, a RWL is a violent upside rejection. RWL is the most common follow-on shape after BS, forming the BS --> RWL chain. When the RWL is particularly strong (rejecting back through the prior BS origin), it signals that the BS was a false breakout and the dominant trend has reversed.

Variant - Piercing Whiplash (PWL): A Whiplash where the initial surge is so powerful it pierces through multiple IP zones before being rejected. The rejection point is typically a Deep Hard Wall or Institutional Magnet from Engine 1. PWL is the Whiplash equivalent of PBS, indicating extreme volatility.

Variant - Piercing Reverse Whiplash (PRWL): A Reverse Whiplash with piercing characteristics. After a BS or PBS, the rejection is so violent it pierces back through the BS origin and continues into prior structural levels. PRWL is the strongest reversal signal in the taxonomy and almost always leads to a new SS phase at a lower (or higher) range, or directly to a Whiplash Storm.

2.5 Whiplash Storm (WLS)

Definition: An extended sequence of rapid, explosive Whiplash moves occurring across multiple sessions. On the 4H timeframe, WLS appears as Swing Scalps with dramatically increased volatility. On every timeframe below 4H, the price action manifests as violent, successive whiplashes in both directions. WLS is essentially SS on the 4H chart but feels like a storm of individual whiplashes on LTFs.

IP Zone Signature: WLS occurs when multiple IP zones of similar pressure degree exist within a narrow range, and price oscillates violently between them. The key identifier is a cluster of 3+ IP zones within 0.30-0.50 DXY points, each with ZFP between 15-40%. Price is caged between these zones but with enough momentum on each swing to produce explosive moves in both directions. The NY Cut Spring Release effect amplifies each oscillation into a full Whiplash on the LTFs.

Duration: Typically 2-5 sessions. WLS almost always resolves into a Breakout Surge when one of the bounding IP zones finally fails under the accumulated pressure. The transition from WLS to BS is one of the highest-confidence forward-mapping predictions in the system because the prolonged oscillation builds institutional positioning pressure that must eventually release directionally.

Variant - Minor Whiplash Storm: A shorter-duration WLS (1-2 sessions) with less extreme volatility. Minor WLS often appears as a transitional shape between BS and SS, effectively serving as a volatile Recovery phase.

3. Shape Classification Decision Matrix

The following matrix provides the mechanical criteria for classifying the current macro shape based on Engine 1 IP zone data. This is not subjective pattern recognition; each classification is determined by quantifiable IP zone metrics produced by Engine 1. The matrix must be applied in sequence: check the highest-priority condition first, and if it matches, assign that shape. If not, proceed to the next condition.

Priority	Condition (IP Zone Metrics)	Shape	Confidence
1	Price breaks through zone with ZFP > 45% AND confirms 15+ min; Measured Move target exists	BS (or PBS if pierces multiple zones)	HIGH
2	Price surges to zone with ZFP < 20%, briefly pierces, then rejects; Liquidity Sweep Target identified	WL (or PWL if multiple zones pierced)	HIGH
3	After BS/WL: price stabilizes at mid-range zone (ZFP 20-40%); duration < 1 session	R (or RS if sharp retracement)	MODERATE
4	3+ IP zones in narrow range (0.30-0.50 DXY); price oscillating violently; 4H shows consolidation	WLS (or Minor WLS if 1-2 sessions)	HIGH
5	Price oscillating between Heavy/Extreme ceiling and floor zones; both ZFP < 30%; duration > 1 session	SS	HIGH
6	After BS: violent reversal in opposite direction through prior origin; Hard Wall rejection at origin	RWL (or PRWL if piercing)	MODERATE-HIGH

Table 2: Macro Shape Classification Decision Matrix

4. Macro Shape Chains

A macro shape chain is a sequence of shapes that occur during a trading week (or across weeks). Chains are not random; they follow deterministic transition rules governed by IP zone mechanics. Each shape transition is triggered by a specific IP zone interaction. The chains observed across 8 months of backtesting are catalogued below, ordered by observed frequency. The most common chain, SS --> BS --> R --> WL, accounts for the majority of observed weeks.

4.1 Observed Chain Catalogue (by Frequency)

Chain	Frequency	Description

SS --> BS --> R --> WL	HIGHEST	The default weekly storyline. Swing Scalps build positioning pressure, a zone fails triggering BS, Recovery consolidates at mid-range, then Whiplash rejects at the target zone. Occurs approximately 2x per week on average (LTF + HTF combined).
SS --> BS	VERY HIGH	Abbreviated version of the default chain. The week ends before R or WL materialize, or R and WL are delayed to the following week. Common on weeks with fewer sessions or reduced volatility triggers.
BS --> R --> RWL	HIGH	Week opens with a BS (often carrying momentum from prior week). Recovery follows, then a Reverse Whiplash rejects the BS move. Signals a false breakout or overextension.
BS --> RWL --> SS --> BS	HIGH	A complete false breakout cycle. Initial BS is rejected by RWL, price consolidates into SS, then a second BS (often in the same direction) succeeds where the first failed. The SS phase serves as a re-accumulation before the genuine breakout.
WL --> RWL --> SS --> BS	MODERATE	Opening Whiplash is reversed, leading to consolidation, then breakout. Often seen at week open when prior week ended at a major IP zone.
R --> RWL --> SS --> BS	MODERATE	Recovery from prior week BS is reversed, leading to new consolidation and eventual breakout.
BS --> RWL --> PBS	MODERATE	Initial BS is reversed, but the reversal triggers an even more powerful Piercing BS. This chain produces the largest weekly ranges.
BS --> R --> PBS	MODERATE	BS followed by Recovery, then an even stronger second leg. PBS pierces through the initial BS target to a deeper Structural Extreme Magnet.
SS --> BS --> SS --> BS	MODERATE	Double consolidation-breakout pattern. Two separate BS moves within one week, each preceded by its own SS phase.
4H BS --> Minor WLS --> SS	MODERATE	HTF breakout creates volatile oscillation on LTFs, then settles into SS. The 4H Breakout Surge is the structural driver; the Minor WLS is the LTF expression.
PBS --> SS (Minor WLS)	MODERATE	Piercing BS is immediately followed by volatile consolidation. The Minor WLS is the market absorbing the shock of the piercing move.
WLS --> BS --> WLS	MODERATE	Whiplash Storm resolves into Breakout Surge, which triggers another WLS at a higher level. This is the violent oscillation regime.
WLS --> SS --> PBS	MODERATE	WLS exhausts into SS consolidation, then a Piercing BS breaks out. The WLS phase is essentially SS on 4H; the PBS is the directional resolution.
WLS --> PBS --> PRWL --> SS	LOW-MODERATE	WLS resolves violently with PBS, which is immediately reversed by PRWL, then settles. High-volatility regime with extreme directional swings.
PBS --> R/SS --> WL --> RWL	LOW-MODERATE	After PBS, Recovery or SS consolidation, then Whiplash and Reverse Whiplash at the target zone.

PBS --> R/SS --> PRWL --> SS	LOW-MODERATE	PBS followed by Recovery, then Piercing Reverse Whiplash rejects the entire PBS move, settling into SS.
WLS --> R --> PBS/WL	LOW	WLS transitions to brief Recovery, then either a Piercing BS continuation or Whiplash reversal.
WLS --> R --> SS	LOW	WLS exhausts into Recovery, then SS. Quiet resolution after violent oscillation.

Table 3: Observed Macro Shape Chains (8-Month Backtest, Ordered by Frequency)

4.2 Transition Rules

Each shape transition is governed by specific IP zone conditions. The transition is deterministic once the IP zone metrics are known. This is the core mechanism that allows forward mapping: by knowing the current shape and the IP zones that price will encounter, the next shape in the chain can be predicted with high confidence.

Current Shape	Next Shape	IP Zone Trigger	Probability
SS	BS	A bounding zone with ZFP > 45% fails with confirmation (15+ min)	HIGH (when ZFP > 60%)
SS	WLS	3+ zones in narrow range; volatility triggers amplify oscillations	MODERATE
BS	R	Price reaches Measured Move target zone (ZFP 20-40%); momentum fades	HIGH
BS	RWL	Price reaches Hard Wall / Institutional Magnet; strong rejection	HIGH (at Hard Walls)
BS	PBS	Zone fails with extreme momentum; target is distant Structural Extreme	MODERATE (at high ZFP)
R	SS	Stabilization persists > 1 session; both bounding zones hold	MODERATE-HIGH
R	WL/RWL	Recovery reaches opposing IP zone; Whiplash rejection	MODERATE
R	BS (2nd leg)	Recovery at a mid-range zone that also fails; cascade continues	MODERATE
WL	RWL	Whiplash rejected at zone; opposite direction momentum builds	MODERATE-HIGH
WL	SS	Whiplash exhausts momentum; price consolidates near rejection point	MODERATE
RWL	SS	Reversal exhausts; new range established	HIGH

RWL	BS (new dir)	Reversal breaks through opposing zone	MODERATE
WLS	BS	A bounding zone of the storm finally fails under pressure	VERY HIGH
WLS	SS	Volatility diminishes; oscillation range narrows	MODERATE
PBS	R/SS	Piercing move exhausts at distant Structural Extreme	HIGH
PBS	PRWL	Piercing move overextends; violent reversal	MODERATE
PRWL	SS	Reversal exhausts; new consolidation range established	HIGH

Table 4: Macro Shape Transition Rules with IP Zone Triggers

5. IP Zone Anchoring: How Engine 1 Zones Determine Engine 2 Shapes

The foundational principle of Engine 2 is that macro shapes are anchored to IP zones. Without IP zone data, shape prediction is speculative. With IP zone data, shape prediction becomes deterministic because each shape is defined by how price interacts with specific zone types. This section details the exact anchoring mechanism for each shape.

5.1 The May 18-22 Case Study

The week of May 18-22, 2026, provides a perfect illustration of IP zone anchoring. Without IP zones mapped, the price action appeared to follow the chain BS --> R --> PBS (Piercing Breakout Surge). The Breakout Surge was followed by a brief Recovery, and it appeared the surge would continue piercing higher. However, this analysis was wrong because the IP zone at 99.43-99.67 was not mapped.

The 99.43-99.67 zone was a Heavy to Extreme pressure resistance zone with high structural confidence (likely a Confluent Ceiling or Repeated High Magnet from Engine 1 Pillar 5 Step 0.5). This zone had ZFP well below 30%, meaning it was highly likely to HOLD against a breakout attempt. The "Recovery" that appeared to be forming was actually price approaching this strong IP resistance zone, and the "PBS" that appeared to be starting was actually the zone rejecting the surge.

Correct chain with IP zones: BS --> Reverse Whiplash --> SS --> Whiplash Storm. The BS reached the 99.43-99.67 zone, was violently rejected (Reverse Whiplash), then price consolidated in a volatile range (SS transitioning to WLS) as the market oscillated between the lower IP zones and the rejected ceiling. Had the IP zone been mapped beforehand, the incorrect BS --> R --> PBS prediction would never have been made, because the ZFP of the 99.43-99.67 zone would have flagged that a breakout was unlikely there.

5.2 Anchoring Rules by Shape

Shape	IP Zone Role	Engine 1 Metrics Required	Forward Mapping Logic
SS	Ceiling and Floor zones define the range	Both bounding zones: ZFP < 30%, Confluence Score >= 6, Regime = Consolidation/Momentum	If both boundary zones hold, SS continues. If one zone ZFP > 45%, prepare for BS transition.
BS	The failing zone triggers BS; target zone determines magnitude	Failing zone: ZFP > 45% with breakout pre-conditions active. Target: Measured Move Projection from Pillar 9 Step 4, anchored to Structural Extreme Magnet	If failing zone ZFP exceeds 45% with 2+ breakout signals, BS is imminent. Target is the nearest Structural Extreme Magnet, not a blind projection.
R	Mid-range zone provides temporary friction	Recovery zone: ZFP 20-40%, typically Post-Expiry Ghost or One-Touch. Not a Hard Wall.	Recovery is temporary. If the next zone in the direction of the prior BS has ZFP < 30%, expect R --> WL/RWL. If the zone has ZFP > 45%, expect R --> BS (second leg).
WL/RWL	Hard Wall or Institutional Magnet rejects the move	Rejection zone: ZFP < 20%, Confluence Score >= 8, Liquidity Sweep Target from Pillar 5 Step 0, or Institutional Magnet from Step 0.5d	Whiplash is guaranteed at Hard Walls with ZFP < 15%. The Liquidity Sweep Zone designation from Engine 1 is the primary flag. Piercing variant if multiple zones are pierced before rejection.
WLS	Cluster of 3+ zones in narrow range creates oscillation cage	Zone cluster: 3+ IP zones within 0.30-0.50 DXY, each ZFP 15-40%, with NY Cut option cage amplifying oscillations	WLS continues until one bounding zone fails. The zone most likely to fail is the one with highest ZFP + most breakout pre-conditions active. WLS --> BS is the highest-confidence transition.
PBS	Multiple intermediate zones fail in sequence; distant Structural Extreme is the target	All intermediate zones: ZFP > 50%. Target: deep Structural Extreme Magnet (Measured Move anchored)	PBS occurs when the failing zone has ZFP > 60% and the Measured Move points to a distant, verified Structural Extreme. Each intermediate zone it pierces adds momentum (cascade effect).
PRWL	Institutional Magnet at depth of reversal provides the rejection anchor	Rejection at a deep Institutional Magnet or Confluent Floor/Ceiling from Pillar 5 Step 0e/0.5d	PRWL occurs after PBS when the target zone is a maximum-confidence structural anchor. The reversal is violent because stops from the entire PBS move are clustered at the target.

Table 5: IP Zone Anchoring Rules by Macro Shape

6. Forward Mapping Procedure

The forward mapping procedure is executed after Engine 1 has produced the DXY IP Zone Map for the target week. It takes the IP Zone Map as input and produces a Macro Shape Chain Map as output. The

procedure is mechanical, deterministic, and auditable at every step.

6.1 Step 1: Current Shape Identification

Identify the macro shape that is currently active as of the Friday close (the temporal boundary for the report). This is the starting shape of the chain. Apply the Shape Classification Decision Matrix (Table 2) to the current price position relative to the IP Zone Map. If price is between two Heavy/Extreme zones with low ZFP, the current shape is SS. If price has just broken through a zone with high ZFP, the current shape is BS. If price is at a Hard Wall showing rejection, the current shape is WL or RWL.

6.2 Step 2: Transition Probability Assessment

For the current shape, consult the Transition Rules (Table 4) to identify all possible next shapes and their IP zone triggers. For each possible transition, evaluate whether the trigger conditions are met by the IP Zone Map:

- If a bounding zone has ZFP > 45% with active breakout pre-conditions, the SS --> BS transition probability is HIGH.
- If the Measured Move target zone is a Hard Wall with ZFP < 20%, the BS --> RWL transition probability is HIGH.
- If a zone cluster of 3+ zones exists in the current range, the SS --> WLS or WLS continuation probability is HIGH.
- If the Measured Move target zone has moderate ZFP (20-40%), the BS --> R transition probability is MODERATE-HIGH.

6.3 Step 3: Chain Extension

Extend the chain forward by applying the transition rules iteratively. For each predicted shape, identify the IP zone that would serve as the trigger for the next transition, and evaluate whether that trigger would be met. Continue until either (a) the end of the trading week is reached, (b) the chain enters SS with no zone having ZFP > 35% (terminal SS), or (c) the chain prediction confidence drops below MODERATE.

Chain Extension Audit Trail (MANDATORY): Each chain extension must document: the current shape, the IP zone being evaluated as a trigger, the specific ZFP and breakout pre-conditions, the predicted next shape, and the confidence level. This audit trail ensures every prediction is independently verifiable from the Engine 1 IP Zone Map data alone.

6.4 Step 4: Chain Confidence Scoring

Each chain prediction receives a confidence score based on the cumulative probability of its transitions. The chain confidence is the product of individual transition confidences, adjusted for the historical frequency of the chain.

Confidence tiers for individual transitions: HIGH = 0.80, MODERATE-HIGH = 0.65, MODERATE = 0.50, LOW-MODERATE = 0.35, LOW = 0.20. For a chain of two transitions (e.g., SS --> BS --> R), the chain confidence is the product of the two individual transition confidences. If either transition has a confidence below MODERATE, the chain prediction is flagged as [LOW CONFIDENCE - consider alternative chains].

6.5 Step 5: Alternative Chain Analysis

For every primary chain prediction, generate at least one alternative chain that would activate if the primary chain fails at a specific transition point. The alternative chain shares the same starting shape but diverges at the first transition where an alternative trigger could activate. The IP zone that determines which chain activates is explicitly identified as the "Decision Zone."

Example from May 18-22: The primary chain was BS --> R --> PBS (incorrect). The alternative chain was BS --> RWL --> SS --> WLS (correct). The Decision Zone was 99.43-99.67. If this zone held (ZFP < 30%, Hard Wall), the alternative chain activated. If this zone failed (ZFP > 45%), the primary chain would have activated. The Decision Zone's actual ZFP and structural classification from Engine 1 determines which chain plays out.

7. Engine Integration: How Engine 1 Feeds Engine 2

Engine 2 does not operate independently. It is built entirely on top of Engine 1's output. Every macro shape, every transition, and every chain prediction is anchored to specific IP zones from Engine 1. The integration follows a strict data flow: Engine 1 produces the IP Zone Map, and Engine 2 consumes it to produce the Shape Chain Map.

7.1 Data Flow

Engine 1 Output	Engine 2 Consumer	Usage
Zone range + Pressure Degree	Shape Classification (Table 2)	Determines whether a zone is a ceiling, floor, or mid-range friction for the current shape
Zone Failure Probability (ZFP)	Transition Rules (Table 4)	Determines whether a zone will hold or fail, triggering a shape transition
Measured Move Projection	BS/PBS target identification	Identifies the target zone for Breakout Surge, anchoring the BS magnitude
Structural Extreme Magnets (Pillar 5 Step 0/0.5)	WL/RWL/PRWL rejection point identification	Identifies the exact zone where Whiplash rejection will occur
Liquidity Sweep Targets (Pillar 5 Step 0)	WL/PWL trigger identification	Identifies zones where brief pierce + rejection (Whiplash) is expected

Breakout pre-conditions (Pillar 9)	BS transition timing	Determines when a zone failure is imminent, triggering SS --> BS
Option expiry timing (Protocol A)	Shape transition timing within sessions	Identifies Spring Release windows that trigger shape transitions at 10:01 AM NY
Regime classification (Pillar 9)	Chain confidence scoring	Consolidation regime supports SS continuation; Breakout regime supports BS transition
Volatility Triggers (Protocols D/E/G)	Catalyst-driven shape transitions	Treasury auctions, ON RRP, EIA shocks can trigger BS or amplify WLS

Table 6: Engine 1 to Engine 2 Data Flow

7.2 Execution Sequence

Engine 2 executes after Engine 1 has completed all 9 Pillars and produced the final IP Zone Map. The execution sequence is as follows:

- Engine 1 completes: Full IP Zone Map with zones, levels, pressure degrees, ZFP, Measured Move Projections, and Structural Extreme Magnets.
- Engine 2 Step 1: Current Shape Identification using the Classification Decision Matrix applied to the IP Zone Map.
- Engine 2 Step 2: Transition Probability Assessment for the current shape using Transition Rules and IP Zone triggers.
- Engine 2 Step 3: Chain Extension by iteratively applying transition rules, with mandatory Chain Extension Audit Trail.
- Engine 2 Step 4: Chain Confidence Scoring based on cumulative transition probabilities and historical chain frequency.
- Engine 2 Step 5: Alternative Chain Analysis identifying Decision Zones and alternative chain paths.
- Output: Macro Shape Chain Map integrated into the final PDF report alongside the IP Zone Map.

8. Macro Shape Chain Map Output Format

The Macro Shape Chain Map is produced as a section within the Leviathan Quant Report PDF, directly following the IP Zone Map section. It presents the forward-mapped shape chain for the coming week with full traceability to the IP zones that anchor each shape and transition.

8.1 Chain Map Header

```
MACRO SHAPE CHAIN MAP | WEEK OF [DATE]
Current Shape: [SHAPE] | Confidence: [TIER]
Primary Chain: [SS] --> [BS] --> [R] --> [WL] | Chain Confidence: [X]%
Alternative Chain: [SS] --> [WLS] --> [BS] | Chain Confidence: [X]%
Decision Zone: [ZONE RANGE] | [PRESSURE DEGREE] | ZFP: [X]%
```

8.2 Shape Detail Blocks

```
SHAPE: [NAME] | Duration: [ESTIMATED SESSIONS]
Anchor Zone: [ZONE RANGE] | Pressure: [DEGREE] ([X]/14) | ZFP: [X]%
IP Zone Role: [Ceiling/Floor/Trigger/Rejection/Target]
Transition Trigger: [Zone failure / Zone rejection / Catalyst / Time expiry]
Next Shape: [PREDICTED] | Confidence: [TIER]
Decision Zone (if alternative exists): [ZONE] | [Condition for alternative]
```

8.3 Chain Extension Audit Trail

```
AUDIT TRAIL:
Step 1: Current = [SS] | Zone [98.15-98.25] ZFP=22% (holds) + Zone
[98.75-98.80] ZFP=52% (fail risk)
--> Prediction: SS --> BS | Confidence: HIGH (ZFP>45% + macro catalyst +
option expiry Friday)
Step 2: Current = [BS] | Target = [99.30] (Repeated High Magnet, Apr 1 + Apr
27) | Measured Move Anchored
--> Prediction: BS --> RWL | Confidence: HIGH (target is Hard Wall, ZFP=12%,
Institutional Magnet)
Step 3: Current = [RWL] | Stabilization zone = [98.15-98.25] ZFP=22%
--> Prediction: RWL --> SS | Confidence: MODERATE-HIGH (stabilization at
mid-range zone)
Chain: SS --> BS --> RWL --> SS | Chain Confidence: 0.80 x 0.80 x 0.65 =
41.6% (MODERATE)
```

9. Zero-Tolerance Rules for Engine 2

Engine 2 inherits all Zero-Tolerance Rules from Engine 1 (v4.3). The following additional rules apply specifically to the Macro Shape Chain Map:

> RULE 1: No Shape Without IP Zone Anchoring. Every macro shape prediction must be anchored to at least one specific IP zone from Engine 1 with a stated ZFP, Pressure Degree, and structural role. A shape prediction without an IP zone anchor is classified as UNANCHORED and may not be included in the primary chain.
>
> RULE 2: No Chain Without Audit Trail. Every chain extension step must document the current shape, the IP zone trigger, the trigger conditions evaluated, the predicted next shape, and the confidence level. A chain without a complete audit trail is classified as UNVERIFIED.
>
> RULE 3: Temporal Boundary Inheritance. Engine 2 inherits the Temporal Boundary Rule from Engine 1. All IP zone data used for chain mapping must be pre-week data. No in-week price action may be used to adjust the chain map during the target week. Intra-week chain validation is permissible for learning purposes only, not for the published report.

RULE 4: No Behavioral Generalizations. The ANTI-SLOP rule from Engine 1 extends to Engine 2. Shape durations, transition probabilities, and chain frequencies must be derived from backtested data, not from behavioral estimates like "typically lasts 2-3 sessions" or "usually extends 0.618 of the range." Every quantitative statement must be traceable to either (a) a specific backtested observation count, or (b) a specific IP zone metric from Engine 1.

RULE 5: Mandatory Alternative Chain. Every primary chain prediction must include at least one alternative chain. The Decision Zone that determines which chain activates must be explicitly identified with its ZFP and the condition under which the alternative chain takes priority.

RULE 6: No Chain Extension Beyond Moderate Confidence. If the cumulative chain confidence drops below MODERATE (below 25% cumulative), the chain must be terminated and the remaining week flagged as [DISCOVERY ZONE - insufficient IP zone data for reliable chain extension]. This prevents low-confidence speculation from being presented as analysis.

RULE 7: WLS Resolution Direction Must Be Justified. When predicting WLS --> BS, the direction of the BS (upside or downside) must be justified by identifying the specific bounding zone most likely to fail, including its ZFP, breakout pre-conditions, and the macro catalyst direction. A WLS --> BS prediction without directional justification is classified as INCOMPLETE.

10. Weekly Execution Protocol

The following protocol defines how Engine 2 is executed as part of the weekly Leviathan Quant Report generation. It adds approximately 2-3 minutes to the total report generation time, bringing the total to approximately 10-11 minutes.

Step	Action	Time	Dependencies
1	Complete Engine 1: Full 9-Pillar 6D IP Zone Map	~8 min	All Engine 1 protocols
2	Identify current macro shape using Classification Matrix	~30 sec	IP Zone Map + prior week shape
3	Evaluate all possible transitions from current shape	~1 min	Transition Rules + IP Zone triggers
4	Extend primary chain with audit trail	~1 min	Transition probabilities + IP Zone metrics
5	Score chain confidence	~15 sec	Cumulative transition probabilities
6	Generate alternative chain(s) with Decision Zones	~1 min	Alternative triggers + IP Zone data
7	Format Chain Map section and integrate into PDF	~30 sec	Chain Map output format
Total		~12 min	

Table 7: Engine 2 Weekly Execution Protocol

11. Shape-to-Zone Classification Quick Reference

This reference table provides a rapid lookup for determining which IP zone classification from Engine 1 corresponds to each macro shape role. This is used during Step 2 (Transition Probability Assessment) to quickly identify which zones in the IP Zone Map are candidates for triggering each possible transition.

Shape Role	Engine 1 Zone Classification	Key ZFP Range	Key Pillar(s)
SS Ceiling	Heavy/Extreme Resistance	< 30%	Pillar 1 (Pivot R), Pillar 8 (Option Wall above), Pillar 5 Step 0.5 (Repeated High Magnet)
SS Floor	Heavy/Extreme Support	< 30%	Pillar 1 (Pivot S), Pillar 5 Step 0 (Liquidity Sweep Target), Pillar 8 (Option Wall below)
BS Trigger Zone	Zone with ZFP > 45%	> 45%	Pillar 9 (Breakout pre-conditions), Pillar 8 (Expiring option wall)
BS Target Zone	Structural Extreme Magnet	N/A (target)	Pillar 5 Step 0/0.5, Pillar 9 Step 4 (Measured Move)
R Stabilization Zone	Moderate/Moderate-Heavy	20-40%	Pillar 1 (mid-range pivot), Post-Expiry Ghost
WL Rejection Zone	Hard Wall / Institutional Magnet	< 15%	Pillar 5 Step 0e/0.5d, Pillar 8 (Critical Option Wall)
WLS Oscillation Cage	3+ zone cluster in 0.30-0.50 range	15-40% each	Pillar 8 (option cage), Pillar 1 (pivot cluster)
PBS Target	Deep Structural Extreme Magnet	N/A (distant target)	Pillar 5 Step 0.5 (3+ date magnet), Pillar 9 Step 4 (anchored projection)
PRWL Rejection Anchor	Maximum-confidence structural anchor	< 10%	Pillar 5 Step 0.5d (Institutional Magnet), Confluent Floor/Ceiling

Table 8: Shape-to-Zone Classification Quick Reference

12. Future Enhancements (Phase 3 Roadmap)

The following enhancements are planned for future phases of Engine 2 development. They are documented here to establish the roadmap but are NOT part of the current v1.0 specification.

12.1 Cross-Pair Shape Chain Correlation

Map macro shape chains across the three traded component pairs (AUD/USD, GBP/USD, EUR/USD) to identify when multiple pairs are simultaneously in BS or WLS phases. Cross-pair BS alignment among AU, GU, and EU is a cascade trigger that amplifies DXY directional moves. This enhancement would add a "Cross-Pair Confluence Score" to each shape transition, increasing confidence when two or more of the traded pairs confirm the same directional move. Non-traded component pairs (USD/JPY, USD/CHF, USD/CAD) remain available as supplementary IP discovery tools but are not included in the Confluence Score calculation.

12.2 Session-Level Shape Mapping

Decompose the weekly macro shape chain into session-level (Asian, London, NY) sub-chains. Each session has distinct volatility characteristics and institutional flow patterns. The Asian session typically manifests SS or Minor WLS; London often triggers BS; NY produces the decisive shape transitions at the 10:01 AM Spring Release. Session-level mapping would allow intra-day tactical execution aligned with the macro shape chain.

12.3 Historical Chain Frequency Database

Build a quantitative database of observed chain frequencies from the 8-month backtest, with chain frequency broken down by market regime (Consolidation, Momentum, Breakout, Regime Change). This would allow chain probability estimates to be regime-conditional rather than unconditional, significantly improving forward mapping accuracy. Current chain frequencies are observational; the database would make them statistically rigorous.

12.4 Machine Learning Shape Recognition

Train a pattern recognition model on the labeled backtest data to automatically identify shape transitions from real-time price action, reducing the lag between a shape transition occurring and it being detected. The model would be trained on the IP zone metrics and price action features that define each shape, not on raw candlestick patterns.